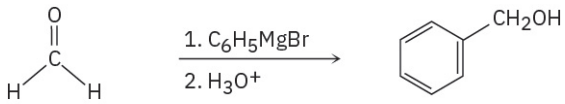
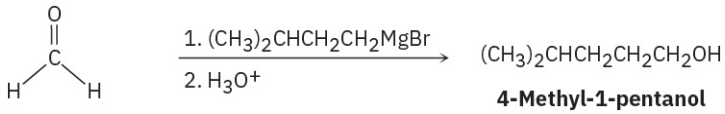
(e)



Benzyl alcohol

(f)



4-Methyl-1-pentanol

PROBLEM 17-11 Cyclohexanone + CH_3CH_2MgBr

PROBLEM 17-12 1. *p*-TosCl, pyridine; 2. NaCN

PROBLEM 17-13 **(a)** 2-Methyl-2-pentene **(b)** 3-Methylcyclohexene **(c)** 1-Methylcyclohexene **(d)** 2,3-Dimethyl-2-pentene **(e)** 2-Methyl-2-pentene

PROBLEM 17-14 **(a)** 1-Phenylethanol **(b)** 2-Methyl-1-propanol **(c)** Cyclopentanol

PROBLEM 17-15 **(a)** Hexanoic acid, hexanal **(b)** 2-Hexanone **(c)** Hexanoic acid, no reaction

PROBLEM 17-16 S_N2 reaction of F^- on silicon with displacement of alkoxide ion.

PROBLEM 17-17 Protonation of 2-methylpropene gives the *tert*-butyl cation, which carries out an electrophilic aromatic substitution reaction.

PROBLEM 17-18 Disappearance of –OH absorption; appearance of C=O

PROBLEM 17-19 **(a)** Singlet **(b)** Doublet **(c)** Triplet **(d)** Doublet **(e)** Doublet **(f)** Singlet

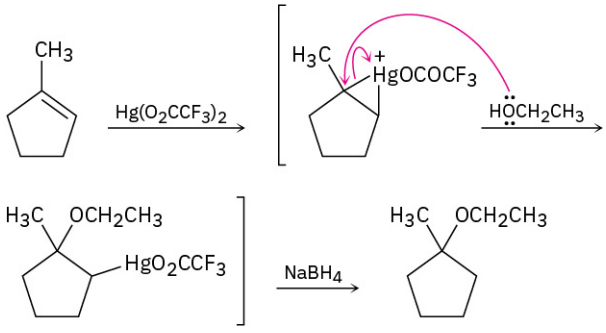
Chapter 18

PROBLEM 18-1 **(a)** Diisopropyl ether **(b)** Cyclopentyl propyl ether **(c)** *p*-Bromoanisole or 4-bromo-1-methoxybenzene **(d)** 1-Methoxycyclohexene **(e)** Ethyl isobutyl ether **(f)** Allyl vinyl ether

PROBLEM 18-2 A mixture of diethyl ether, dipropyl ether, and ethyl propyl ether is formed in a 1 : 1 : 2 ratio.

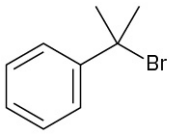
PROBLEM 18-3 **(a)** $CH_3CH_2CH_2O^-$ + CH_3Br **(b)** PhO^- + CH_3Br **(c)** $(CH_3)_2CHO^-$ + $PhCH_2Br$ **(d)** $(CH_3)_3CCH_2O^-$ + CH_3CH_2Br

PROBLEM 18-4



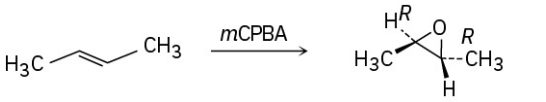
PROBLEM 18-5 **(a)** Either method **(b)** Williamson **(c)** Alkoxymercuration **(d)** Williamson

PROBLEM 18-6 **(a)** Bromoethane > 2-Bromopropane > Bromobenzene **(b)** Bromoethane > Chloroethane > 1-Iodopropene

PROBLEM 18-7 **(a)**  **(b)** 

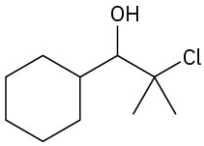
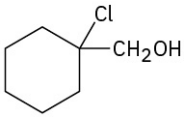
PROBLEM 18-8 Protonation of the oxygen atom, followed by E1 reaction

PROBLEM 18-9 Br^- and I^- are better nucleophiles than Cl^-.

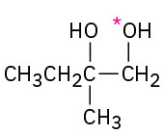
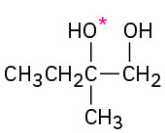
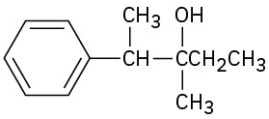
PROBLEM 18-10 

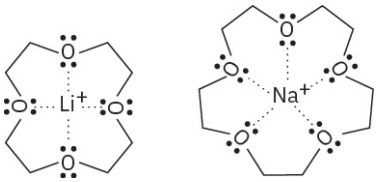
trans-2-Butene trans-2,3-Epoxybutane

PROBLEM 18-11 Epoxidation of *cis*-2-butene yields *cis*-2,3-epoxybutane, while epoxidation of *trans*-2-butene yields *trans*-2,3-epoxybutane.

PROBLEM 18-12 **(a)**  **(b)** 

PROBLEM 18-13 **(a)** 1-Methylcyclohexene + OsO_4; then $NaHSO_3$ **(b)** 1-Methylcyclohexene + *m*-chloroperoxybenzoic acid, then H_3O^+

PROBLEM 18-14 **(a)**  **(b)**  **(c)** 

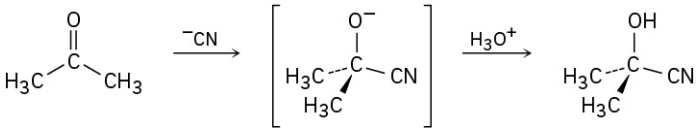
PROBLEM 18-15 

PROBLEM 18-16 **(a)** 2-Butanethiol **(b)** 2,2,6-Trimethyl-4-heptanethiol **(c)** 2-Cyclopentene-1-thiol **(d)** Ethyl isopropyl sulfide **(e)** *o*-Di(methylthio)benzene **(f)** 3-(Ethylthio)cyclohexanone

PROBLEM 18-17 **(a)** 1. $LiAlH_4$; 2. PBr_3; 3. $(H_2N)_2C{=}S$; 4. H_2O, NaOH **(b)** 1. HBr; 2. $(H_2N)_2C{=}S$; 3. H_2O, NaOH

PROBLEM 18-18 1,2-Epoxybutane

PROBLEM 18-69 Acetyl chloride is more electrophilic than acetone.

PROBLEM 18-70 

PROBLEM 18-71 **(a)** Nucleophilic acyl substitution **(b)** Nucleophilic addition **(c)** Carbonyl condensation

Chapter 19

PROBLEM 19-1 **(a)** 2-Methyl-3-pentanone **(b)** 3-Phenylpropanal **(c)** 2,6-Octanedione **(d)** *trans*-2-Methylcyclohexanecarbaldehyde **(e)** 4-Hexenal **(f)** *cis*-2,5-Dimethylcyclohexanone